Q7. Why weren't the newly observed transitions seen before?

Mattera et al., *Phys. Rev. C* **112**, 064326 (2025)

The newly observed transitions

Four γ lines are flagged with footnote "b" (*newly observed*) in Table I on p. 5. One of these (166 keV) had previously been *proposed* by Hassan et al. (1986) but was never adopted into ENSDF, so the abstract calls only three of them "not previously reported."

E_γ (keV)	Transition $E_i \to E_f$	$J_i^\pi \to J_f^\pi$	I_γ (rel.)	Shown in	Reference (paper)
166.1	2516 → 2350	$4^+ \to 5^+$	0.019(1)	Fig. 4 (p. 6)	p. 3, col. 2
644.5	2547 → 1903	$1^+ \to 0^+$	0.0023(2)	Fig. 6 (p. 6)	p. 3, col. 2
982.8	3395 → 2412	$(4^-) \to 3^+$	0.009(2)	Fig. 7 (p. 6)	p. 3, col. 2
1009.7	3474 → 2464	$3^- \to 3^-$	0.008(1)	Fig. 8 (p. 7)	p. 4–5

Intensities are normalized to $I_\gamma(1596) = 100$. For scale: the strongest transition is **5,000–50,000 times** brighter than these new lines.

Where the paper answers "why weren't these seen before?"

The decisive passage is on **p. 3, column 2**:

From the paper

> *"The ENSDF evaluation of ^{140}La is primarily based on the study by Chand et al. from 1991, which used a limited two-detector setup. Since the work by Chand, there have been considerable advancements in the instrumentation for γ-ray spectroscopy."*

That's the whole answer in one sentence. The rest of this writeup unpacks *why* a two-detector setup in 1991 physically could not see these lines, while 48 detectors in 2025 can.

The pedagogical explanation (for an undergrad)

Imagine you're trying to identify very faint sounds at a crowded dinner party. A weak whisper ($I_\gamma \sim 0.002$) is completely drowned out by the loud conversations ($I_\gamma = 100$). There are three reasons the 2025 experiment could hear the whispers and the 1991 experiment could not.

Reason 1 — Vastly more coincidence statistics

Key Concept

A "singles" spectrum records every γ ray that hits any detector. A "coincidence" spectrum records *pairs* of γ rays that arrive within nanoseconds of each other — i.e., from the same nuclear decay. Requiring coincidence throws away uncorrelated background (cosmics, room activity, random combinations) but costs statistics, because you need *two* detectors to fire for every event you keep.

A bit of math

With N detectors, the number of coincidence pairs is $\binom{N}{2} = N(N-1)/2$:

$N = 2$: 1 pair $\quad$ $N = 10$ (CLARION2) : 45 pairs $\quad$ $N = 48$ (Gammasphere) : 1128 pairs.

So Gammasphere collects **roughly 1000× more coincidence data per decay** than the 1991 setup.

The paper quantifies this directly in Sec. II (p. 2–3):

- **CLARION2**: 1.4×10^{10} singles events, 1.7×10^{7} γ-γ coincidence events.
- **Gammasphere**: 1.9×10^{10} singles events, 2.5×10^{9} γ-γ coincidence events.

Chand et al. (1991) had two detectors. They simply could not accumulate enough γ-γ coincidence pairs to see these lines above the statistical noise, no matter how long they counted.

Reason 2 — Coincidence gating cleans the spectrum

Key Concept

Gating means: select only events where one γ had a specific energy (say 816 keV), and look at the spectrum of the *other* γ in those events. Every peak in the gated spectrum is a transition that was in cascade with the gating transition. Uncorrelated background is suppressed by orders of magnitude.

In a singles spectrum, a weak line at $I_\gamma = 0.009$ sits under the Compton continuum from the three dominant transitions (1596, 487, 816 keV) and is statistically indistinguishable from background. In a gated spectrum, all of that Compton background is *gone* — it wasn't in coincidence with the gate.

Concrete example (Fig. 7, p. 6): the new 983 keV transition depopulates the 3395 keV level down to 2412 keV. The authors gate on the two known transitions *out* of 2412 keV (the 816 keV and 329 keV lines). Any decay that went 3395 → 2412 → lower must have emitted all three γ's. In the gated spectrum, the 983 keV peak pops out cleanly. In a singles spectrum, it's invisible.

Same logic powers Fig. 4 (166 keV confirmed by gating on 242 and 267 from the 2350 level), Fig. 6 (645 keV confirmed by gating on 307 out of the 1903 level), and Fig. 8 (1010 keV confirmed by gating on 868 out of the 2464 level).

Reason 3 — Compton suppression and sensitivity

Both arrays wrap every HPGe crystal in a **BGO veto shield**. When a γ deposits only part of its energy in the HPGe and Compton-scatters out, the BGO catches the scattered photon and the event is rejected. This removes the broad Compton continuum that would otherwise bury weak peaks.

The paper quantifies the resulting sensitivity on **p. 3, col. 2**:

From the paper

"We were able to determine new γ-ray transitions with an intensity as low as 0.000015 *times the strongest transition."*

That's $I_\gamma \sim 1.5 \times 10^{-5}$ relative to the 1596 keV line. For reference, the 1991 work reported nothing weaker than $I_\gamma \sim 10^{-2}$ (Table I). The modern arrays are about **three orders of magnitude more sensitive** in the weak-transition regime.

Reason 4 (bonus) — Nearby contaminants that needed resolving

One subtlety worth flagging because it appears in Fig. 8: near the new 1010 keV peak is the **single-escape peak of the 1596 keV** γ at $1596 - 511 = 1085$ keV. (When a high-energy γ pair-produces in the HPGe and one of the 511 keV annihilation photons escapes the crystal, you get a peak 511 keV below the full-energy peak.) In a 1991 singles spectrum, anything in this energy region sits on the shoulder of a much larger structure. Only a coincidence gate that excludes the 1596 cascade can reveal a weak peak at 1010.

Tying it together

Bottom line

The new transitions weren't seen in 1991 because they are 10^3–10^4 times weaker than the main lines, and a two-detector setup lacks the coincidence statistics, background suppression, and Compton rejection needed to resolve peaks at that level. The 2025 measurement wins on three axes simultaneously:

1. $\sim 10^3\times$ **more coincidence pairs** from the $N(N-1)/2$ scaling with detector count.
2. **Coincidence gating** that kills uncorrelated Compton background.
3. **BGO Compton suppression** that kills partial-energy events at the detector level.

The point isn't that Chand et al. did a bad job in 1991 — they did an excellent job for the instrumentation available. The point is that a 4π Compton-suppressed array is a qualitatively different tool, and reaching $I_\gamma \sim 10^{-5}$ simply wasn't possible 30 years ago.

Where to point in the paper if asked

- **p. 2–3, Sec. II (Experiment)**: detector counts, event statistics, efficiency calibration — the quantitative comparison to a 1991 setup.
- **p. 3, col. 2**: explicit sentence identifying Chand et al. (1991) as a two-detector measurement and naming the sensitivity limit (1.5×10^{-5}).
- **Table I, p. 5–6**: the four transitions with footnote "b."

- **Figs. 4, 6, 7, 8 (p. 6–7)**: the actual gated-spectrum evidence for each new line. Each is a before/after demonstration of why gating is decisive.